Albert Quizon

21. 2D Dumbell

DAE Approach

- 2 masses, 1 rod
- 2D Dumbell spinning

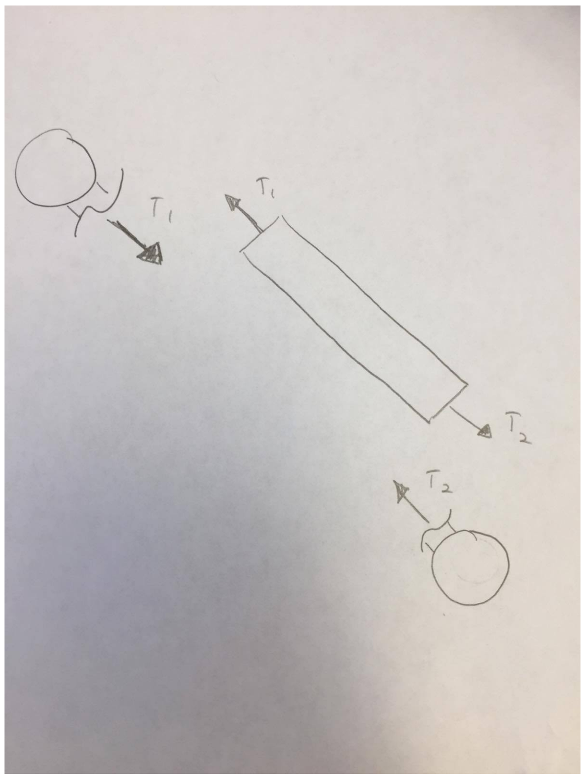

Figure 1: FBD for 2D Dumbell.

Make a cut at the end of the rod. Rod is massless, so two tension forces will be equal and opposite.

Constraint (with derivatives)

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 = l^2 \tag{1}$$

$$2(x_1 - x_1)(\dot{x}_2 - \dot{x}_1) - 2(y_2 - y_1)(\dot{y}_2 - \dot{y}_1) = 0 \tag{2}$$

$$(x_2 - x_1)(\ddot{x}_2 - \ddot{x}_1) - (y_2 - y_1)(\ddot{y}_2 - \ddot{y}_1) = (\dot{y}_2 - \dot{y}_1)^2 - (\dot{x}_2 - \dot{x}_1)^2 \tag{3}$$

2 LMBs

$$
\begin{aligned}
F &= m(\ddot{x}\hat{i} + \ddot{y}\hat{j}) && (4)\\
F\hat{i} &= m(\ddot{x}\hat{i} + \ddot{y}\hat{j}) \cdot \hat{i} && (5)\\
F\hat{j} &= m(\ddot{x}\hat{i} + \ddot{y}\hat{j}) \cdot \hat{j} && (6)\\
F\hat{i} &= m\ddot{x} && (7)\\
F\hat{j} &= m\ddot{y} && (8)\\
l &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} && (9)\\
m_1\ddot{x}_1 &= T\frac{x_2 - x_1}{l} && (10)\\
m_1\ddot{y}_1 &= T\frac{y_2 - y_1}{l} && (11)\\
m_2\ddot{x}_2 &= T\frac{x_1 - x_2}{l} && (12)\\
m_2\ddot{y}_2 &= T\frac{y_1 - y_2}{l} && (13)
\end{aligned}
$$

Write Eq (7),(8),(9),(10), and (3) in matrix form

$$
\begin{bmatrix}
m_1 & 0 & 0 & 0 & -\frac{x_2-x_1}{l}\\
0 & m_1 & 0 & 0 & -\frac{y_2-y_1}{l}\\
0 & 0 & m_2 & 0 & -\frac{x_1-x_2}{l}\\
0 & 0 & 0 & m_2 & -\frac{y_1-y_2}{l}\\
x_1 - x_2 & y_2 - y_1 & x_2 - x_1 & y_1 - y_2 & 0
\end{bmatrix}
\begin{bmatrix}
\ddot{x}_1\\ \ddot{y}_1\\ \ddot{x}_2\\ \ddot{y}_2\\ T
\end{bmatrix}
=
\begin{bmatrix}
0\\ 0\\ 0\\ 0\\ (\dot{y}_2 - \dot{y}_1)^2 - (\dot{x}_2 - \dot{x}_1)^2
\end{bmatrix}
$$

Above matrix equation for RHS file.

Plots comparing tolerances of 1e-10 and 1e-2. Initial conditions:

$$
\begin{bmatrix}
x_1\\ y_1\\ x_2\\ y_2\\ vx_1\\ vy_1\\ vx_2\\ vy_2
\end{bmatrix}
=
\begin{bmatrix}
-1\\ 0\\ 1\\ 0\\ 0\\ -0.5\\ 0\\ 0.5
\end{bmatrix}
$$

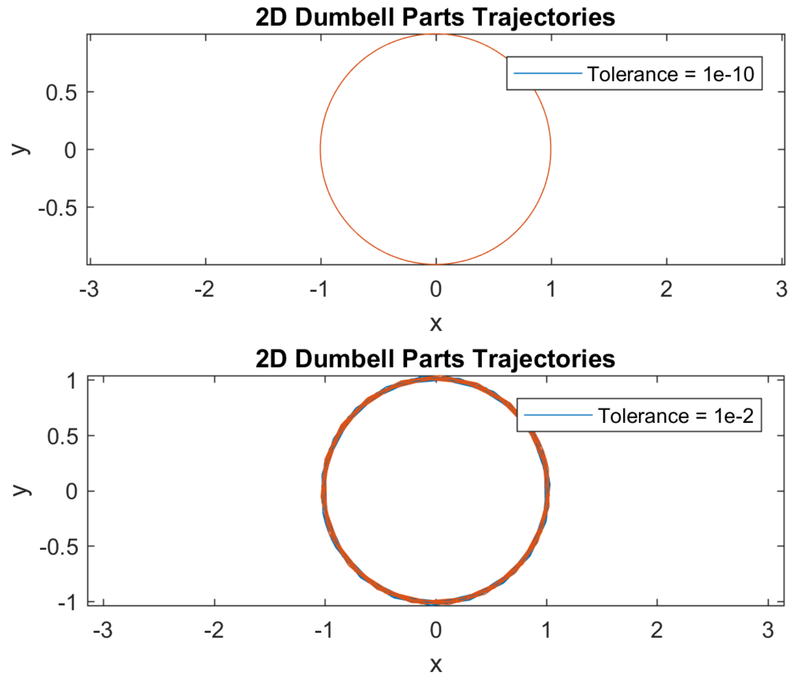

Figure 2: Trajectory of Dumbell Parts.

Error in Kinetic Energy

$$\epsilon_{KE} \quad = \quad \left| \frac{KE_{expected} - KE_{num\ sol}}{KE_{expected}} \right|$$

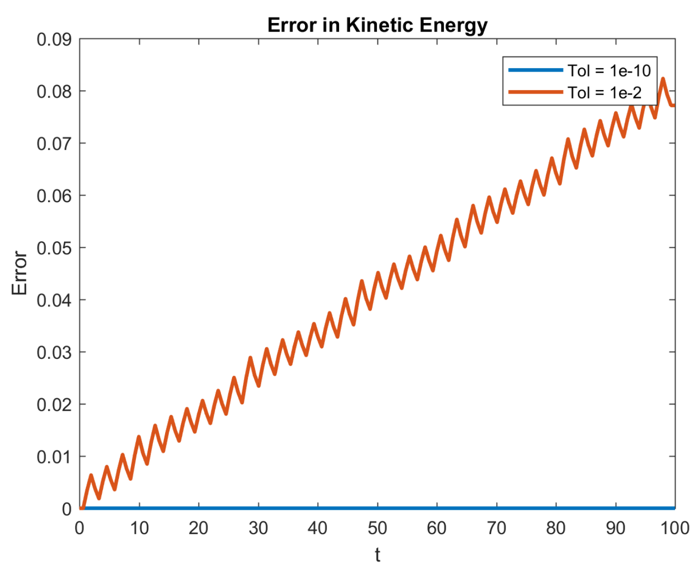

Figure 3: Error in Dumbell Kinetic Energy.

Error in Constraint Equation

Initial conditions: $[x0_1,\ x0_2,\ y0_1,\ y0_2]$

$$\begin{aligned} l &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ l_{expected} &= \sqrt{(x0_2 - x0_1)^2 + (y0_2 - y0_1)^2} \\ \epsilon_{cons} &= \left| \frac{l_{expected} - l_{num\ sol}}{l_{expected}} \right| \end{aligned}$$

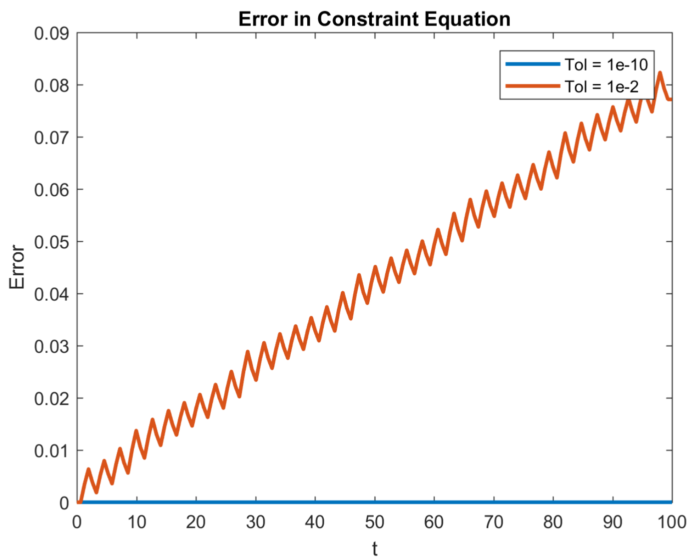

Figure 4: Error in Dumbell Constraint Equation.

In general, error in kinetic energy and in the constraint equation are on the order of the relative and absolute tolerances.

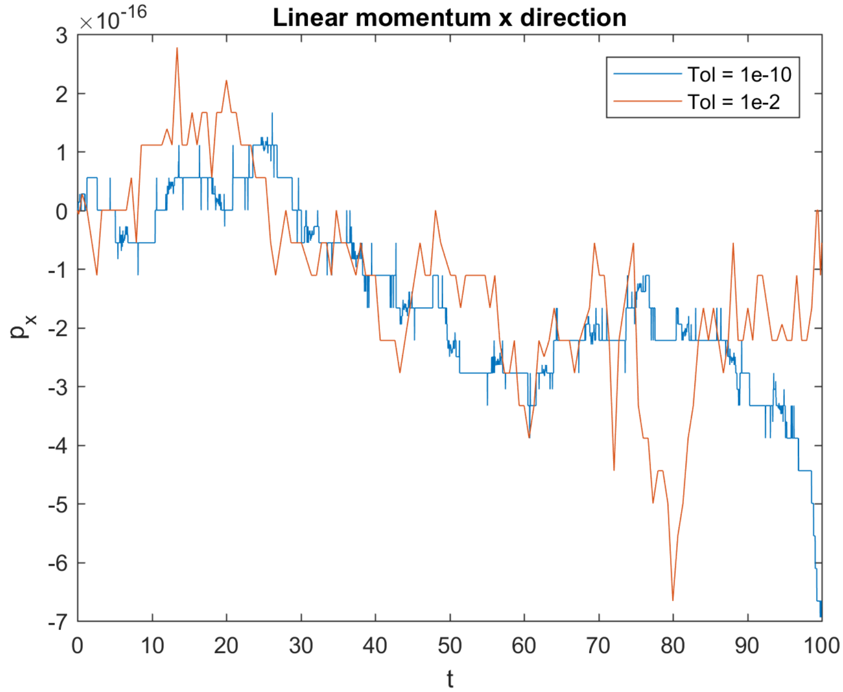

Figure 5: Dumbell Linear Momentum x direction.

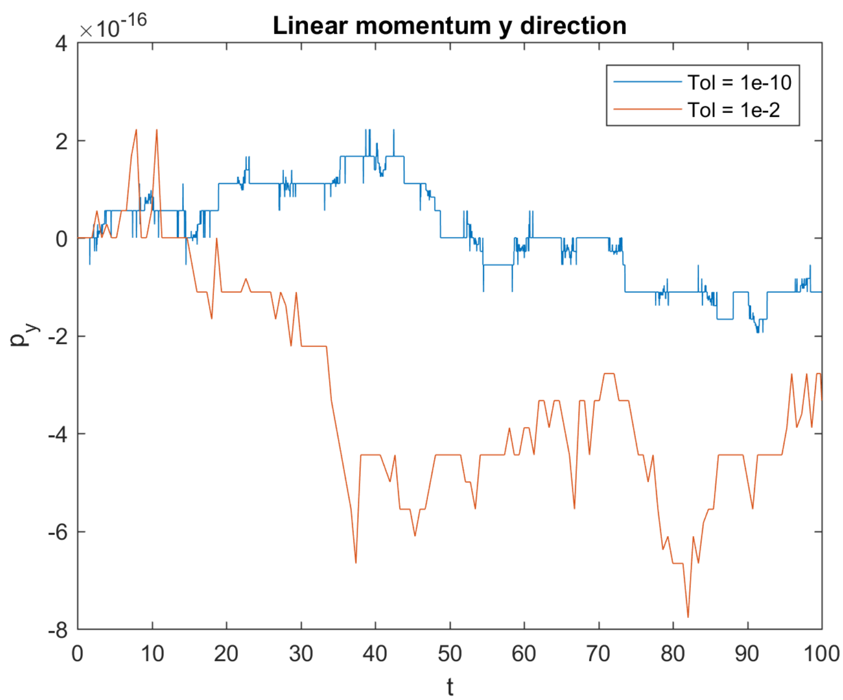

Figure 6: Dumbell Linear Momentum y direction.

The difference in linear momentum differs on the order of 1e-16, which is much smaller than the set tolerances.

The initial conditions were changed so the system was rotated by 45^o, using sin and cos factors.

$$\begin{bmatrix} x_1 \\ y_1 \\ x_2 \\ y_2 \\ vx_1 \\ vy_1 \\ vx_2 \\ vy_2 \end{bmatrix} = \begin{bmatrix} -1cos(\frac{\pi}{4}) \\ -1sin(\frac{\pi}{4}) \\ 1cos(\frac{\pi}{4}) \\ 1sin(\frac{\pi}{4}) \\ 0.5sin(\frac{\pi}{4}) \\ -0.5cos(\frac{\pi}{4}) \\ -0.5sin(\frac{\pi}{4}) \\ 0.5cos(\frac{\pi}{4}) \end{bmatrix}$$

Plug in values into M matrix, obtain singular matrix, so should be no solution. However, in the following plots, sin and cos were used which introduce numerical errors on the order of 1e-15, so we can still obtain some sort of solution.

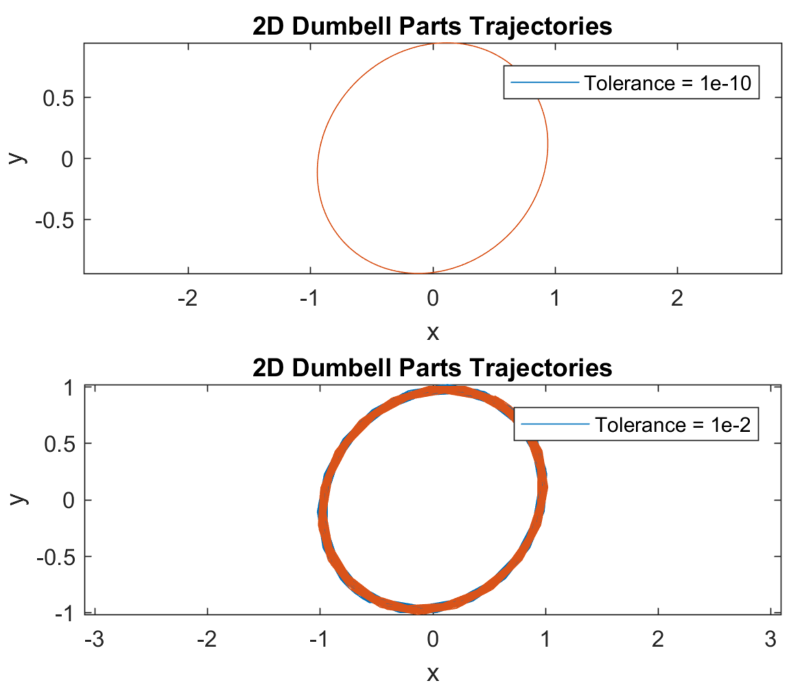

Figure 7: Trajectory of Dumbell Parts Rotated.

Trajectory of parts becomes more of an oval due to numerical imprecision when using sin and cos.

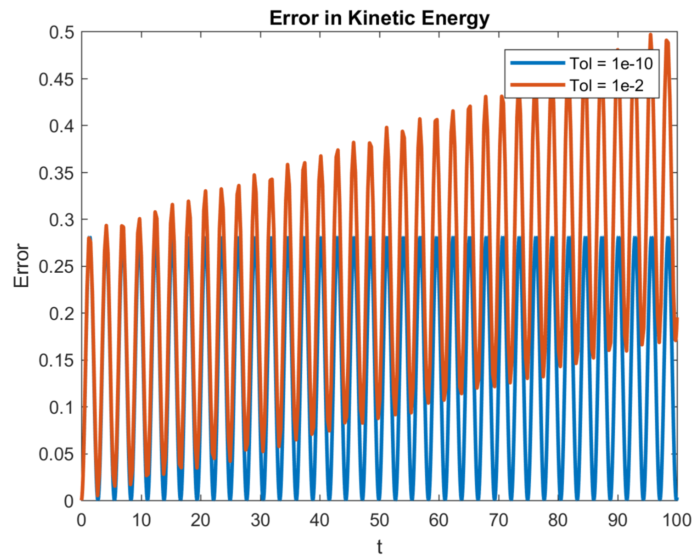

Figure 8: Error in Dumbell Kinetic Energy (Rotated IC).

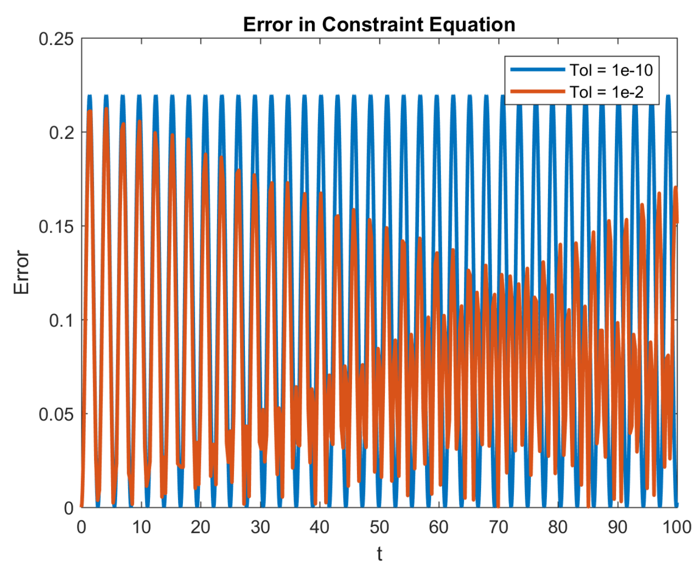

Figure 9: Error in Dumbell Constraint Equation (Rotated IC).

In these cases, the error in both the kinetic energy and the constraint equation are on a much larger order of magnitude than before. This is due to initial numerical error when calculating $\sin(\pi/4)$ and $\cos(\pi/4)$.

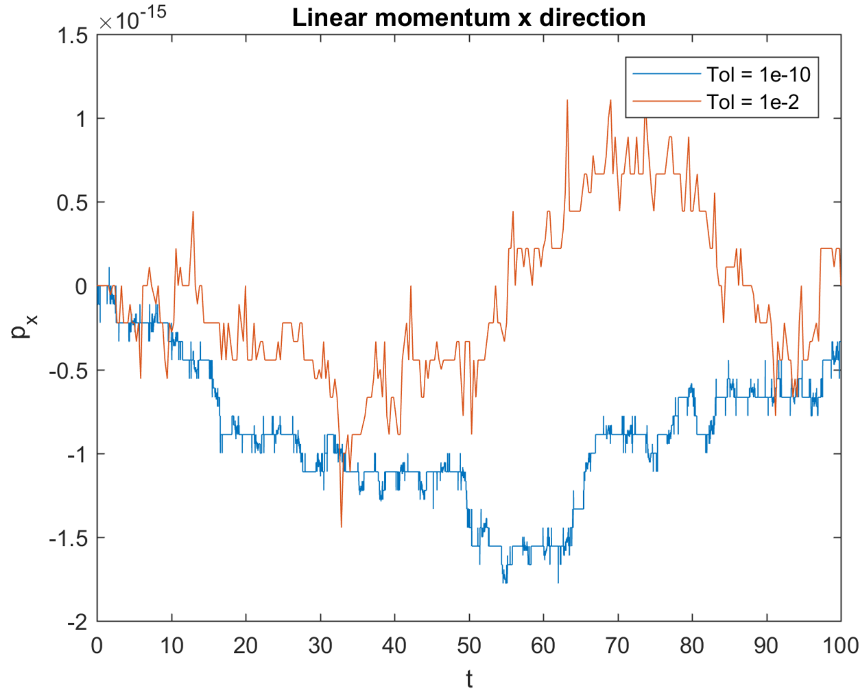

Figure 10: Dumbell Linear Momentum x direction (Rotated IC).

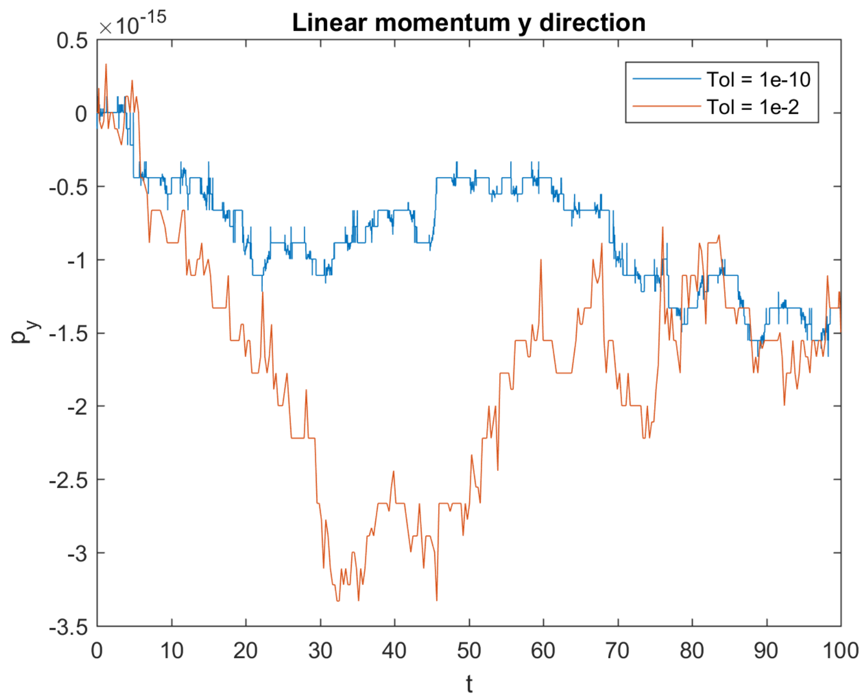

Figure 11: Dumbell Linear Momentum y direction (Rotated IC).

The difference in linear momentum still differs on the order of 1e-16, which is much smaller than the set tolerances.

Code

```
function z_dot = RHS_Dumbell(t,z,p)
% RHS_Dumbell: 1st order form of ODE
%
%   INPUTS
%       t           time in sec
%       z           z array of stored x values
%       p           parameters array
%
%   OUTPUTS
%       z_dot       derivative of z
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%   Cornell University
%   MAE 5730: Intermediate Dynamics and Vibrations

% z = [x,y,z,xdot,ydot,zdot]

x1 = z(1);
y1 = z(2);
x2 = z(3);
y2 = z(4);
x1dot = z(5);
y1dot = z(6);
x2dot = z(7);
y2dot = z(8);

l = sqrt((x2-x1)^2 + (y2-y1)^2);    % calculation of length

% From matrix equation
M = [p.m1, 0, 0, 0, -(x2 - x1)/l;...
    0, p.m1, 0, 0, -(y2-y1)/l;...
    0, 0, p.m2, 0, -(x1-x2)/l;...
    0, 0, 0, p.m2, -(y1-y2)/l;...
    (x1-x2), (y2-y1), (x2-x1), (y1-y2), 0];
b = [0;0;0;0;(y2dot-y1dot)^2 - (x2dot - x1dot)^2];
RHS = M\b;

x1ddot = -RHS(1);
y1ddot = -RHS(2);
x2ddot = -RHS(3);
y2ddot = -RHS(4);
```

```
z_dot = [x1dot;y1dot;x2dot;y2dot;x1ddot;y1ddot;x2ddot;y2ddot];
end

%   2D Dumbell
%
%   Albert Quizon
%   Cornell University
%   MAE 5730: Intermediate Dynamics and Vibrations

% Parameters
p.m1 = 1;
p.m2 = 1;
l = 2;
v = 0.5;

% Initial Conditions
theta = 0;
x1 = -l/2*cos(theta);
x2 = l/2*cos(theta);
y1 = -l/2*sin(theta);
y2 = l/2*sin(theta);
vx1 = v*sin(theta);
vx2 = -v*sin(theta);
vy1 = -v*cos(theta);
vy2 = v*cos(theta);

z_0 = [x1,y1,x2,y2,vx1,vy1,vx2,vy2];

tspan = [0, 100]; % initial and final times

options = odeset('RelTol', 1e-10, 'AbsTol', 1e-10);
options2 = odeset('RelTol', 1e-2, 'AbsTol', 1e-2);
[t,z] = ode45(@RHS_Dumbell,tspan,z_0,options, p);
[t2,z2] = ode45(@RHS_Dumbell,tspan,z_0,options2, p);

% Trajectory
figure(1)
subplot(2,1,1)
plot(z(:,1), z(:,2), z(:,3), z(:,4))
title('2D Dumbell Parts Trajectories')
xlabel('x')
ylabel('y')
axis('equal')
legend('Tolerance = 1e-10')
subplot(2,1,2)
plot(z2(:,1), z2(:,2), z2(:,3), z2(:,4))
title('2D Dumbell Parts Trajectories')
xlabel('x')
```

```
ylabel('y')
axis('equal')
legend('Tolerance = 1e-2')

% Conservation of momentum
figure(2)
px = p.m1*z(:,5) + p.m2*z(:,7);
py = p.m1*z(:,6) + p.m2*z(:,8);
px2 = p.m1*z2(:,5) + p.m2*z2(:,7);
py2 = p.m1*z2(:,6) + p.m2*z2(:,8);
plot(t,px,t2,px2);
title('Linear momentum x direction')
xlabel('t')
ylabel('p_x')
legend('Tol = 1e-10', 'Tol = 1e-2')

figure(3)
px = p.m1*z(:,5) + p.m2*z(:,7);
py = p.m1*z(:,6) + p.m2*z(:,8);
px2 = p.m1*z2(:,5) + p.m2*z2(:,7);
py2 = p.m1*z2(:,6) + p.m2*z2(:,8);
plot(t,py,t2,py2);
title('Linear momentum y direction')
xlabel('t')
ylabel('p_y')
legend('Tol = 1e-10', 'Tol = 1e-2')

% Energy
figure(5)
energy = 0.5*p.m1*(z(:,5).^2 + z(:,6).^2) +
 0.5*p.m2*(z(:,7).^2 + z(:,8).^2);
energy_expected = 0.5*p.m1*(vx1^2 + vy1^2) +
 0.5*p.m2*(vx2^2 + vy2^2)*ones(length(t),1);
energy2 = 0.5*p.m1*(z2(:,5).^2 + z2(:,6).^2) +
 0.5*p.m2*(z2(:,7).^2 + z2(:,8).^2);
energy_expected2 = 0.5*p.m1*(vx1^2 + vy1^2) +
 0.5*p.m2*(vx2^2 + vy2^2)*ones(length(t2),1);
error1 = abs(energy-energy_expected)./energy_expected;
error2 = abs(energy2-energy_expected2)./energy_expected2;
plot(t,error1,t2,error2, 'linewidth', 2)
title('Error in Kinetic Energy')
xlabel('t')
ylabel('Error')
legend('Tol = 1e-10', 'Tol = 1e-2')

% Constraint
figure (4)
cons_expected = ones(length(t),1)*((x2-x1)^2 + (y2-y1)^2);
```

```
cons = (z(:,3)-z(:,1)).^2 + (z(:,4) - z(:,2)).^2;
cons_expected2 = ones(length(t2),1)*((x2-x1)^2 + (y2-y1)^2);
cons2 = (z2(:,3)-z2(:,1)).^2 + (z2(:,4) - z2(:,2)).^2;
error_cons = abs(cons-cons_expected)./cons_expected;
error_cons2 = abs(cons2-cons_expected2)./cons_expected2;
plot(t,error_cons, t2,error_cons2, 'linewidth', 2)
title('Error in Constraint Equation')
xlabel('t')
ylabel('Error')
legend('Tol = 1e-10', 'Tol = 1e-2')
```